

FedADAS: Communication-Efficient Federated Distillation for On-Device Driver Yawn Recognition in Vehicular Networks



PAPER

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Motivation

- Driver fatigue is a leading cause of accidents on roads.
- In 2017, US National Highway Traffic Safety Administration reported **91,000 crashes** involved drowsy drivers [1], estimating **50,000 injured** and **800 deaths**.
- In 2019, e-survey of road users' attitudes reports that Austria has the highest driver fatigue percentage (**31.8%**) in Europe [2].

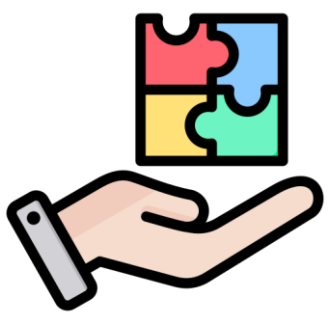
Problem Statement

1. **On-Device Learning:** Driver-monitoring models trained offline on static datasets adapt poorly to real-world driving.
2. **Federated learning struggles in vehicular networks:**
 - Assumes common model architecture; no model heterogeneity.
 - High communication costs and sub-optimal performance under non-IID data.
 - Privacy issues; prone to model inversion attacks.

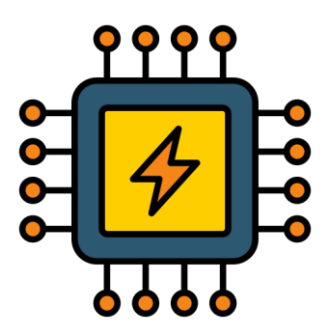
Key Contributions



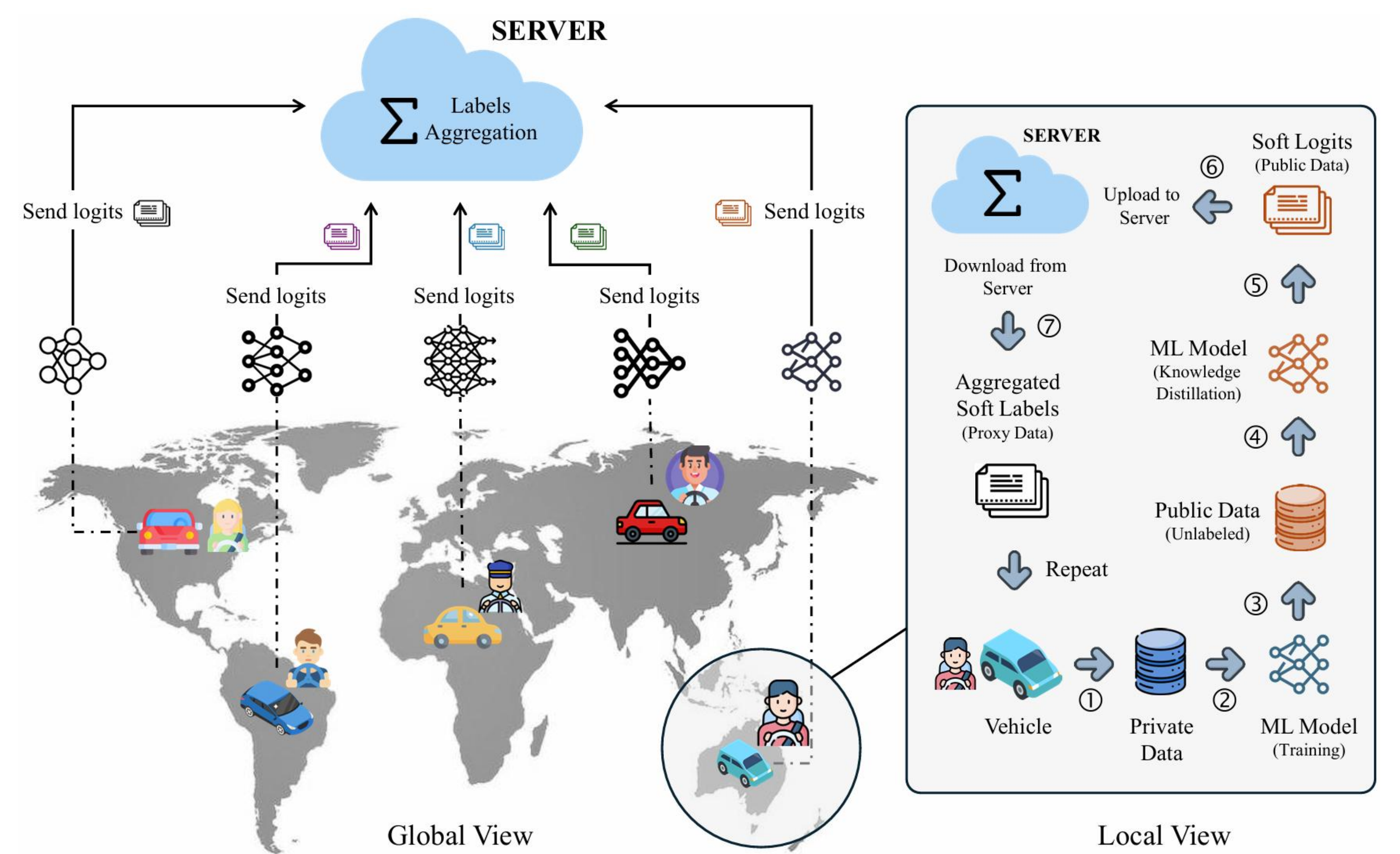
Up to 9974x Lower Communication Cost: Vehicles send only model logits instead of full model parameters (FedAvg).



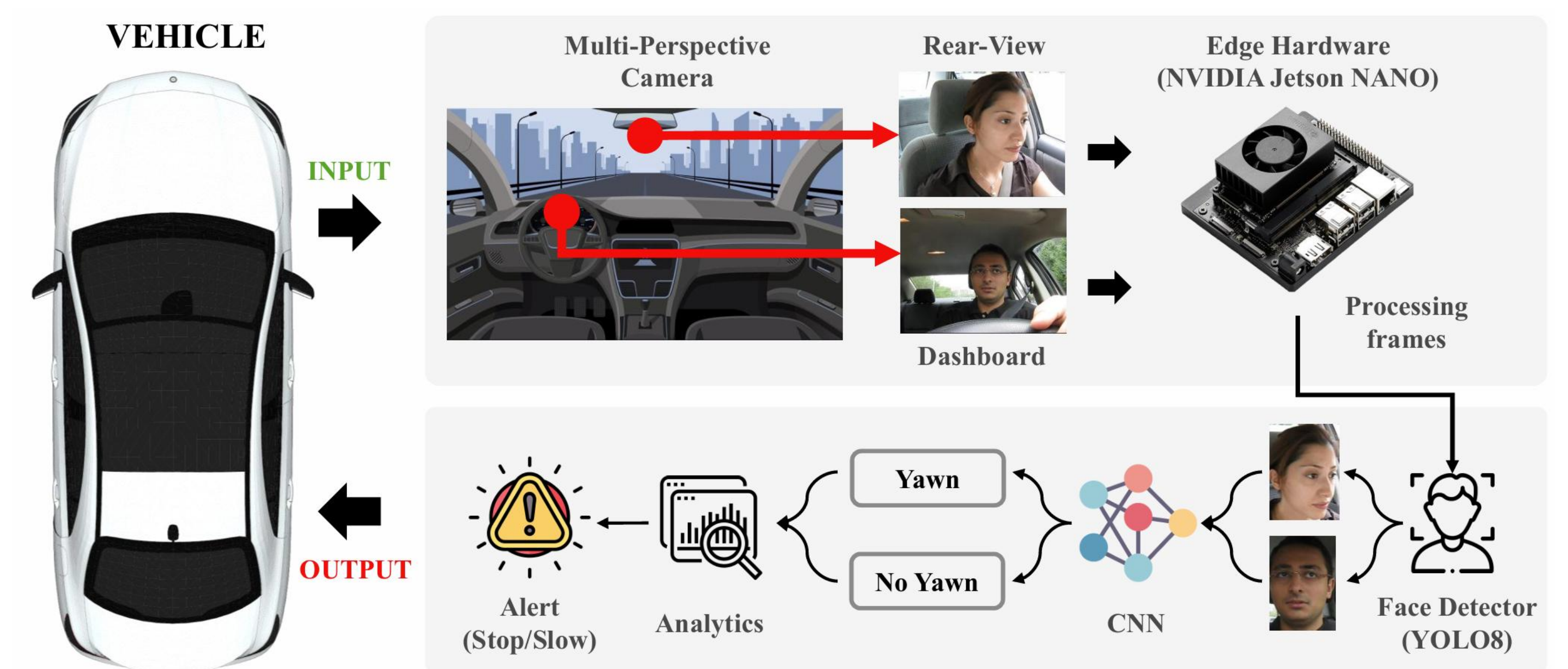
Full Model Heterogeneity: Every vehicle runs a different architecture tuned to its own edge hardware (no shared backbone).



On-device Learning: Proposed two robust models (ME-Net and PE-Net) for on-device learning on edge devices.

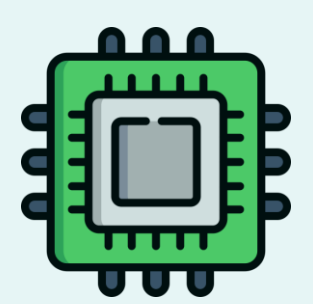


FedADAS Framework



On-Device Driver Yawn Recognition Pipeline

On-Device Learning without pruning or quantization



ME-Net

Memory-Efficient

- ★ Depthwise + pointwise convolutions
- ★ Model size: **0.6 MB**
- ★ Accuracy: **99.30%**
- ★ F1-score: **97.99%**
- ★ **3.81ms** on Jetson Nano
- ★ Best overall training/memory efficiency on edge hardware



PE-Net

Performance-Efficient

- ★ Normal convolutions with large parameter space
- ★ Model size: **99.7 MB**
- ★ Accuracy: **99.39%**
- ★ F1-score: **98.25%**
- ★ **1.99ms** on Jetson Nano
- ★ Best accuracy and fastest inference

FedADAS

FedADAS can train ME-Net, PE-Net, and other architectures together without requiring a shared common backbone.

Conclusion: FedADAS allow vehicles with **fully heterogeneous** model architectures to learn collaboratively **at scale** by exchanging soft predictions only, reducing communication by up to **9974x** compared with FedAvg, while enabling **on-device learning** on edge devices.



References

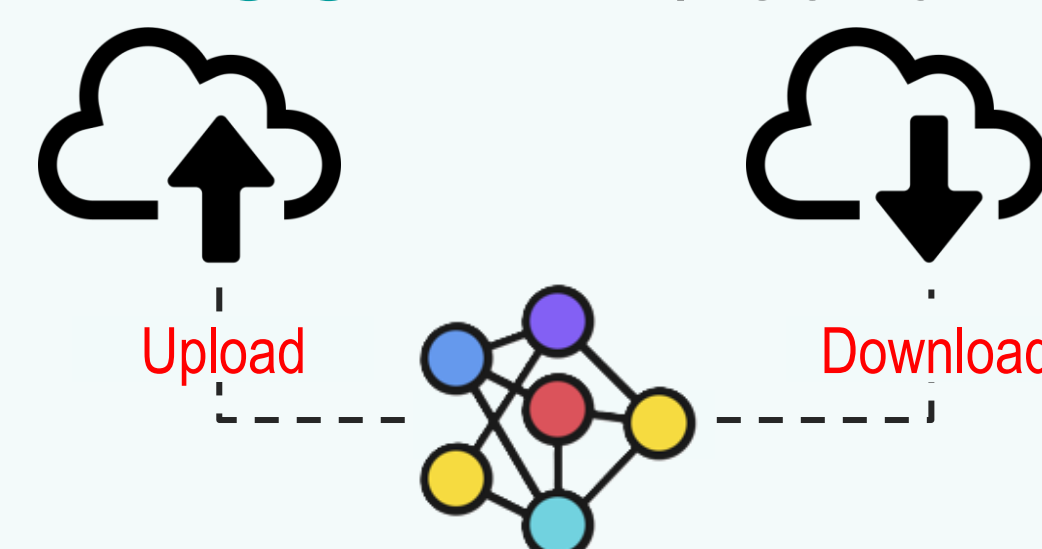
- [1] <https://www.nhtsa.gov/risky-driving/drowsy-driving>
- [2] https://road-safety.transport.ec.europa.eu/system/files/2021-07/road_safety_thematic_report_fatigue_tc_final.pdf

CODE

Performance Comparison – FedADAS vs. FedAvg

FedAvg – PE-Net

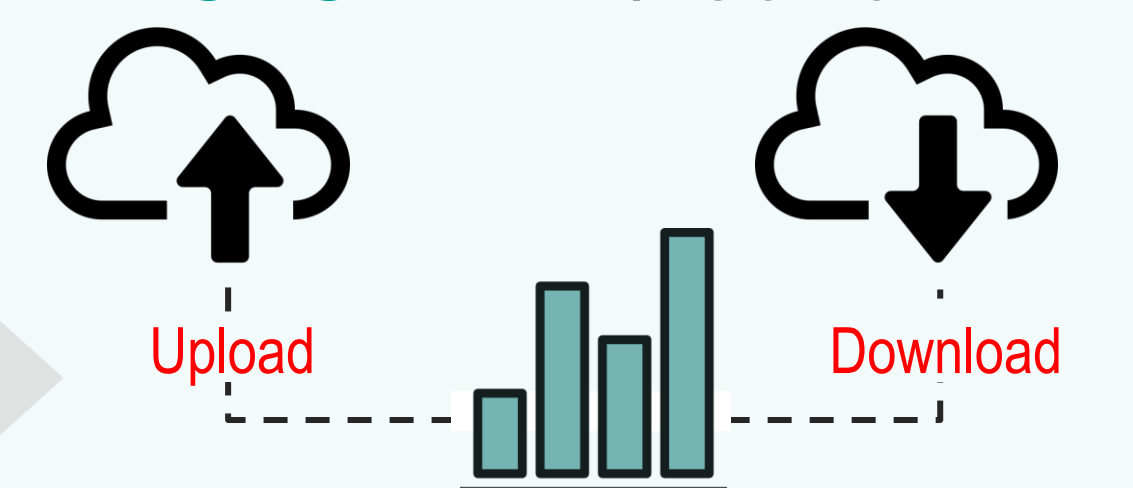
199.4 MB/round



- ★ Full model parameters exchange.
- ★ Scale linearly with model size.

FedADAS – PE-Net

0.02 MB/round



- ★ Only model logits are exchanged.
- ★ Scale linearly with model logits.

76.35%
personalization

76.35%
generalization

+10.83 pp

+21.88 pp

87.18%
personalization

98.23%
generalization

- ★ FedAvg struggles as the number of clients increases
- ★ Latency on Jetson NANO per epoch:
 - ❖ Local Training: $14.26 \pm 0.11s$
- ★ No distillation overhead
- ★ High communication cost
- ★ No model heterogeneity

- ★ FedADAS outperforms FedAvg at scale ($n=115$)
- ★ Latency on Jetson NANO per epoch:
 - ❖ Local Training: $13.85 \pm 0.53s$
 - ❖ Distillation: $1.17 \pm 0.53s$
- ★ Negligible distillation overhead
- ★ **Lower communication cost**
- ★ **With full model heterogeneity**

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